

1. Project Proposal

1.1 Project Introduction

This project aims to develop a comprehensive e-commerce platform that mimics the fundamental functions of leading marketplaces. The platform will provide users with a seamless shopping experience, focusing on displaying products and presenting detailed information about them, in addition to facilitating the process of adding products to the shopping cart.

The general objective is to facilitate the online shopping experience for users by providing an intuitive and effective user interface.

1.2 Project Objectives

The following objectives are set to ensure the success of the project and achieve its vision:

- **Design a user interface (UI) and user experience (UX) that is intuitive and easy to use:** Ensuring the platform is easy to use and visually appealing to users from different technical levels.
- **Full responsiveness:** The platform must work efficiently and adapt to different screen sizes (desktops, tablets, smartphones).
- **Optimal performance and loading speed:** Improving platform performance to ensure fast loading times for pages and a smooth user experience.
- **Effective product display:** Providing a system for displaying products that allows for easy browsing and presents clear and detailed information for each product.
- **Shopping cart management:** Enabling users to add products to the shopping cart, modify quantities, and remove them easily.

1.3 Project Scope

This section defines the project scope clearly, including in-scope and out-of-scope features and functionalities, to ensure focus and avoid scope creep.

In-Scope Features:

- **User Registration/Login Page:** Will be implemented using Local Storage instead of a full backend authentication system. Users can register, log in, and manage their session locally within the browser, without server-side authentication or database integration.
- **Homepage:** Displaying featured products and main categories.
- **Product Listing Pages:** Browsing products by category or search.
- **Product Detail Page:** Displaying product images, description, price, and specifications.
- **Shopping Cart:** Adding/removing products, modifying quantities, and total display.
- **Mock Checkout Process:** Simulating payment steps without actual payment processing.
- **Basic Content Management System (CMS):** For adding and modifying products (for administrators).

Out-of-Scope Features:

- **Real Payment Gateway Integration:** The platform will not be integrated with real payment gateways for processing financial transactions.
- **Product Reviews and Ratings System:** A system for product reviews and ratings will not be included.
- **Shipping and Delivery Management System:** A system for managing shipping and delivery will not be implemented.
- **Advanced Admin Dashboard:** The advanced control panel will only be for product management in the basic version.
- **Localization:** The site will only operate in the basic language in the current version.

2. Project Plan

2.1 Timeline & Milestones

The following is an estimated timeline for the main phases of the project. This schedule may be adjusted based on actual progress and any challenges that may arise.

Phase	Suggested Duration	Key Tasks
Week 1	UI/UX Design	- Design user interface (UI/UX) for main pages - Prepare wireframes & mockups
Week 2	HTML & CSS Development	- Build structure using HTML5 - Style components with CSS3 & Flexbox/Grid - Ensure responsive design for various devices
Week 3	JavaScript & API Integration	- Implement dynamic functionality with JavaScript - Fetch and handle data from APIs - Manage shopping cart logic and state
Week 4	Integration & Testing	- Connect front-end with back-end services - Conduct comprehensive functional testing - Bug Fixing
Week 5	Review & Final Delivery	- Review project with supervisors - Prepare final project documentation - Final project delivery

2.2 Key Milestones

- **Week 1 Milestone:** All wireframes and UI mockups approved by the team.
- **Week 2 Milestone:** Fully responsive static pages completed for all major views.
- **Week 3 Milestone:** Dynamic JavaScript functionality and API integration complete.
- **Week 4 Milestone:** All features integrated and tested; critical bugs resolved.
- **Week 5 Milestone:** Final documentation submitted and project delivered to supervisors.

2.3 Resource Allocation

- **GitHub** — Version control and code collaboration.
- **Figma** — UI/UX planning and wireframing.

- **Trello** — Task management and sprint tracking.
- **VS Code** — Primary development environment.
- **HTML5, CSS3, JavaScript (or React)** — Core front-end technologies.

3. Task Assignment & Roles

3.1 Team Member Responsibilities

Each team member has been assigned a clearly defined ownership area to ensure accountability and parallel development:

Team Member	Responsibility
Member 1	Home page + Navigation bar
Member 2	Product Listing Page
Member 3	Cart & Checkout UI
Member 4	User Authentication Pages (Login/Signup)
Member 5	Testing, Documentation, GitHub Integration

3.2 Front-End Task Breakdown

- Design responsive UI using HTML, CSS, JavaScript (or React if used).
- Implement product cards and categories.
- Develop shopping cart interface.
- Create login/signup pages.
- Test responsiveness on mobile and desktop.

3.3 Collaboration Workflow

- Each member works on a dedicated GitHub branch to minimize merge conflicts.
- Pull requests require peer review before merging to the main branch.
- Shared Figma workspace ensures consistent design language across all pages.
- Weekly standups track individual progress and surface blockers early.

4. Risk Assessment & Mitigation Plan

4.1 Risk Register

The table below identifies potential risks, their probability and impact, and the planned mitigation strategy for each:

Risk	Probability	Impact	Solution
Delay in task completion	Medium	High	Set deadlines for each task
Merge conflicts in GitHub	Medium	Medium	Use branches for each member
Inconsistent UI design	High	Medium	Use shared style guide
Missing deadlines	Medium	High	Weekly progress meetings
Bugs in front-end components	High	Medium	Continuous testing

4.2 Mitigation Plan

To reduce risks:

- Use GitHub branches for each member.
- Hold weekly team meetings.
- Use Trello/Jira for tracking tasks.
- Follow common UI design guidelines.
- Test each feature before integration.

5. Key Performance Indicators (KPIs)

The following KPIs will be used to measure the success of the project across performance, quality, and delivery dimensions:

KPI	Target	Measurement Method
Page Load Time	< 3 seconds	Browser Dev Tools / Lighthouse
System Uptime	>= 99%	Monitoring during testing phase
UI Responsiveness	Works on all screen sizes	Cross-device manual testing
Feature Completion Rate	100% of in-scope features	Task tracker (Trello/Jira)
Bug Resolution Time	< 24 hours per critical bug	GitHub Issues log
User Authentication Success	>= 95% success rate	Local Storage login testing
Cart Functionality Accuracy	100% correct total calculation	Functional testing checklist
On-Time Delivery	All 5 phases on schedule	Weekly milestone review

5.1 KPI Review Process

- KPIs will be reviewed at the end of each weekly phase.
- Any KPI falling below target triggers an immediate team review and corrective action.
- Final KPI assessment will be included in the Week 5 project documentation.